Maths Teacher, 5B

Vasant Valley School, New Delhi

Subject: Maths

3139/M5B/0007

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Maths Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Maths are:

Number sense, related concepts and basic number competency
Applications in daily life: word/visual problems

Great job nurturing these talents!

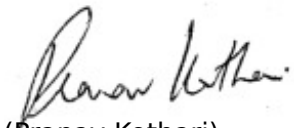
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Maths :

Arithmetic operations: four basic operations, properties
Applications in daily life: money, time, calendar, length etc.

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	C	C	D	C	C	B	C	B	D	C	B	D	A	C	C	B	D	D	C	C	D	D	A	B	A	D	A	C	A	C	D	D	D	D	B	A	C	A	D	D				
1	Aarna Kanoria		B	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	B	✓	C	✓	✓	✓	✓	✓	A	33	668	92	
2	Advitya Arora		✓	✓	C	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	C	✓	✓	C	✓	✓	✓	✓	✓	✓	35	711	96	
3	Agastya Narain		✓	✓	✓	✓	✓	✓	✓	D	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	D	✓	✓	✓	✓	✓	C	A	B	✓	D	C	B	✓	✓	C	29	611	83	
4	Anaheera Singh		✓	D	A	✓	B	✓	✓	D	A	A	D	✓	✓	B	✓	A	✓	C	✓	B	B	✓	✓	✓	B	✓	D	D	✓	✓	A	B	C	✓	D	C	B	✓	✓	C	18	493	49	
5	Anushka		B	A	C	A	B	C	A	C	A	B	D	B	D	✓	D	A	C	B	B	D	B	C	✓	D	✓	B	C	A	B	B	A	B	A	C	D	B	D	B	C	✓	4	328	0	
6	Aradhya		✓	✓	✓	B	D	✓		C	C	D	A	C	D	B	✓	C	A	C	D	✓	C	C	D	D	✓	✓	B	A	C	B	C	✓	C	B	✓	✓	✓	D	✓	A	13	430	23	
7	Atulya P Pandey		✓	A	C	✓	B	✓	✓	D	✓	✓	C	✓	B	✓	✓	C	✓	B	A	✓	✓	A	B	D	C	✓	B	D	B	B	✓	A	✓	✓	✓	B	B	D	C	C	17	481	45	
8	Aveer Kakar		✓	✓	B	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	A	✓	C	✓	C	✓	✓	✓	C	33	666	92	
9	Freya Kapoor		✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	C	✓	C	A	✓	✓	✓	33	675	93	
10	Gaurav Chand		✓	✓	A	B	A	C	✓	✓	✓	A	C	B	D	A	B	C	B	✓	✓	B	✓	B	C	A	C	B	D	B	✓	D	C	A	✓	✓	✓	C	A	D	B	C	12	417	18	
11	Gavish		✓	✓	✓	✓	D	✓	*	C	C	D	A	C	D	B	✓	C	A	C	D	*	C	C	D	D	✓	✓	✓	✓	B	D	C	B	*	✓	C	B	B	*	A	C	11	423	20	
12	Harshit Kumar		✓	✓	*	B	✓	✓	✓	✓	✓	✓	D	✓	C	✓	✓	✓	✓	B	D	D	✓	C	✓	✓	✓	✓	✓	✓	B	B	✓	C	✓	✓	✓	C	B	✓	B	C	25	543	67	
13	Ibrahim Sheikh		✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	B	B	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	34	677	93	
14	Jananya Prasada		✓	B	✓	✓	✓	✓	A	C	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	C	A	✓	C	B	30	603	82	
15	Kabir Krishnan		✓	✓	✓	✓	✓	✓	D	D	✓	A	✓	✓	✓	✓	✓	✓	✓	A	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	A	✓	C	✓	D	A	D	C	B	27	579	76	
16	Krishna Khullar		✓	B	B	A	B	✓	D	D	✓	✓	✓	B	✓	D	A	✓	✓	✓	A	D	B	A	C	C	D	✓	C	✓	✓	A	C	A	✓	C	✓	B	B	✓	C	C	15	478	44	
17	Lila K Gandhi		B	✓	✓	✓	✓	✓	B	✓	✓	A	✓	✓	✓	✓	✓	A	✓	C	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	C	A	✓	✓	✓	✓	✓	D	✓	A	29	609	83	
18	Maya Bolaria		D	✓	C	✓	✓	A	A	D	A	A	C	✓	✓	✓	A	A	C	B	✓	A	A	A	✓	D	C	✓	D	B	✓	D	A	A	B	✓	D	D	B	D	✓	A	12	442	28	
19	Meera S Kanwar		✓	✓	A	✓	✓	✓	✓	C	✓	✓	A	✓	✓	B	B	A	✓	✓	A	D	✓	A	C	✓	C	✓	✓	✓	✓	B	✓	C	✓	✓	A	B	✓	✓	B	C	23	539	65	
20	Nayra Jairath		✓	✓	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	38	790	99	O
21	Priya Kumari		✓	✓	A	✓	✓	✓	A	A	A	D	D	✓	C	A	✓	A	✓	A	A	✓	✓		B	✓	C	A	✓	D	C	A	C	✓	B	B	A	C	D	D	B	A	13	421	19	
22	Reha Allagh		✓	D	A	✓	✓	✓	A	C	✓	✓	D	✓	✓	✓	B	C	✓	C	✓	D	✓	✓	C	✓	B	✓	B	✓	✓	B	B	C	✓	B	✓	✓	B	✓	✓	A	22	527	61	
23	Riana Upadhyay		✓	✓	B	✓		✓	✓	D	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	34	685	94	
24	Saavya Tokas		✓	✓	C	✓	✓	*	✓	*	✓	A	✓	C	✓	✓	✓	✓	✓	C	A	D	C	✓	✓	C	✓	✓	B	D	✓	B	A	A	✓	C	✓	C	A	C	✓	✓	21	528	62	
25	Samara Kaicker		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	B	B	A	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	33	668	92	
26	Shiven Pershad		✓	✓	A	✓	✓	✓	✓	C	✓	A	✓	✓	✓	✓	A	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	D	C	A	✓	✓	✓	✓	B	✓	✓	✓	30	628	86	
27	Simran Bhalla		✓	✓	✓	✓	✓	✓	✓	C	B	A	C	✓	C	✓	B	A	C	B	A	D	✓	A	✓	✓	✓	✓	✓	D	✓	✓	C	A	C	B	C	D	B	C	✓	✓	19	502	53	
28	Subodh Rana		D	A	A	B	B	A	✓	A	B	B	C	B	✓	D	✓	C	B	A	B	B	C	✓	D	A	D	C	B	D	✓	D	C	✓	✓	C	D	✓	B	C	C	✓	9	398	11	
29	Tabbsum Khatun		✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	B	C	✓	A	A	C	C	A	D	✓	✓	D	D	B	B	✓	A	✓	✓	✓	✓	A	C	A	D	✓	B	A	*	19	502	53	
30	Taria Agarwal		B	✓	A	✓	✓	✓	A	✓	✓	✓	✓	A	D	✓	A	✓	✓	✓	A	✓	✓	✓	D	✓	B	✓	✓	✓	✓	B	C	✓	C	✓	✓	C	A	✓	✓	✓	26	561	72	

% answered correct (section)	78	78	44	78	69	81	63	31	75	50	59	72	69	75	66	50	69	44	53	44	69	63	63	66	59	84	63	66	78	28	38	38	66	56	69	38	38	56	63	41	559.3/500			
% answered correct (school students)	80	79	63	72	73	87	68	37	73	58	61	74	76	71	67	56	68	48	51	54	54	60	69	60	67	78	62	69	79	38	56	39	66	63	65	45	49	58	65	54	579.1/500			
% answered correct (all students)	60	66	49	51	59	79	56	36	50	35	47	64	62	52	62	44	58	36	45	49	48	44	54	34	43	67	48	44	66	40	45	29	53	59	54	35	35	47	52	45	506.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**C**	**C**	**D**	**C**	**C**	**B**	**C**	**B**	**D**	**C**	**B**	**D**	**A**	**C**	**C**	**B**	**D**	**D**	**C**	**C**	**D**	**D**	**A**	**B**	**A**	**D**	**A**	**C**	**A**	**C**	**D**	**D**	**D**	**D**	**B**	**A**	**C**	**A**	**D**	**D**				
31	Viir D Singh		B	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	C	✓	A	✓	✓	✓	✓	✓	✓	B	✓	✓	A	C	B	✓	✓	✓	✓	✓	D	✓	C	29	608	83	
32	Vir K Khanna		✓	✓	C	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	37	706	96	

% answered correct (section)	78	78	44	78	69	81	63	31	75	50	59	72	69	75	66	50	69	44	53	44	69	63	63	66	59	84	63	66	78	28	38	38	66	56	69	38	38	56	63	41	559.3/500			
% answered correct (school students)	80	79	63	72	73	87	68	37	73	58	61	74	76	71	67	56	68	48	51	54	54	60	69	60	67	78	62	69	79	38	56	39	66	63	65	45	49	58	65	54	579.1/500			
% answered correct (all students)	60	66	49	51	59	79	56	36	50	35	47	64	62	52	62	44	58	36	45	49	48	44	54	34	43	67	48	44	66	40	45	29	53	59	54	35	35	47	52	45	506.7/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. * Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 5 B, has performed better than the national average in the Maths ASSET paper this year.

Class 5	Maths	No. of students
5A	117%	34
5B (Your Class)	112%	32
5C	118%	32
All sections combined	116%	98

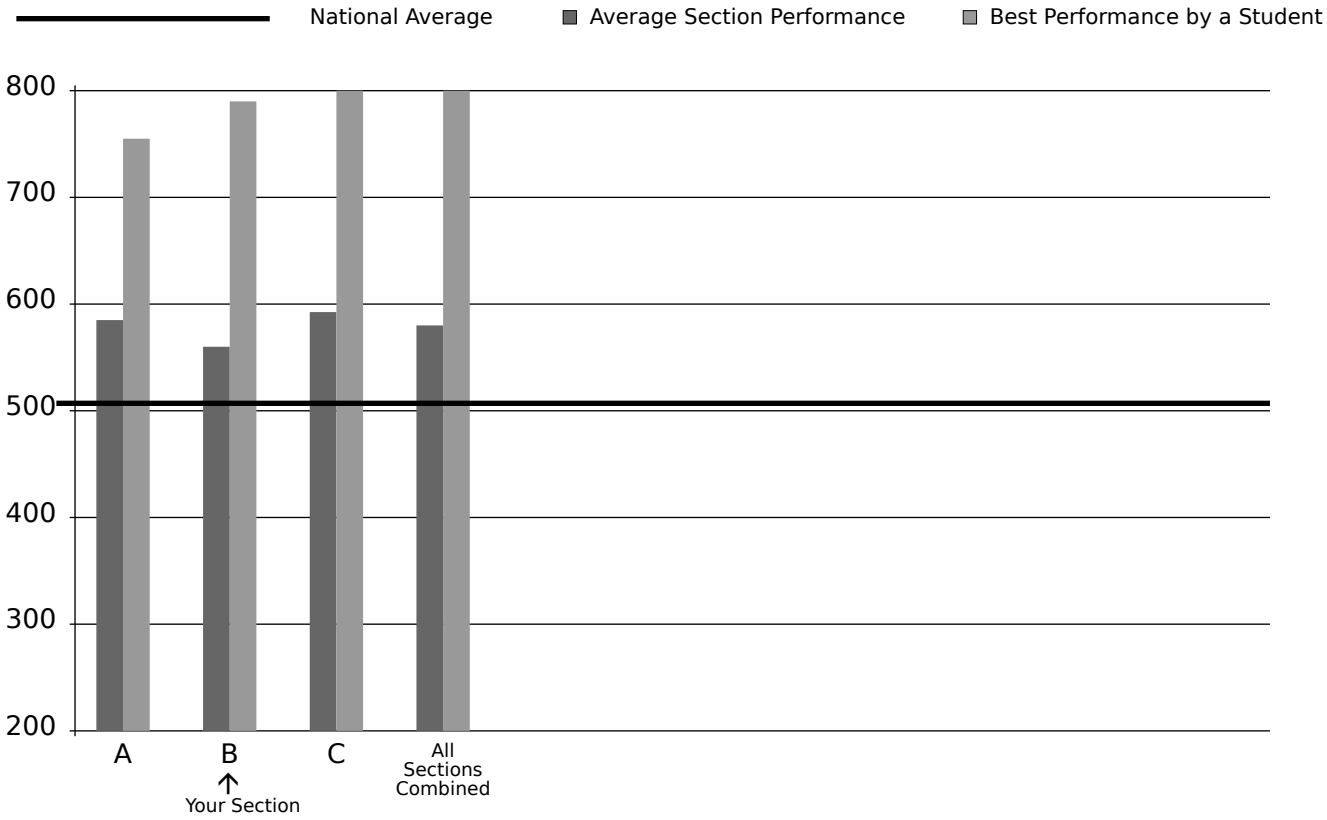
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

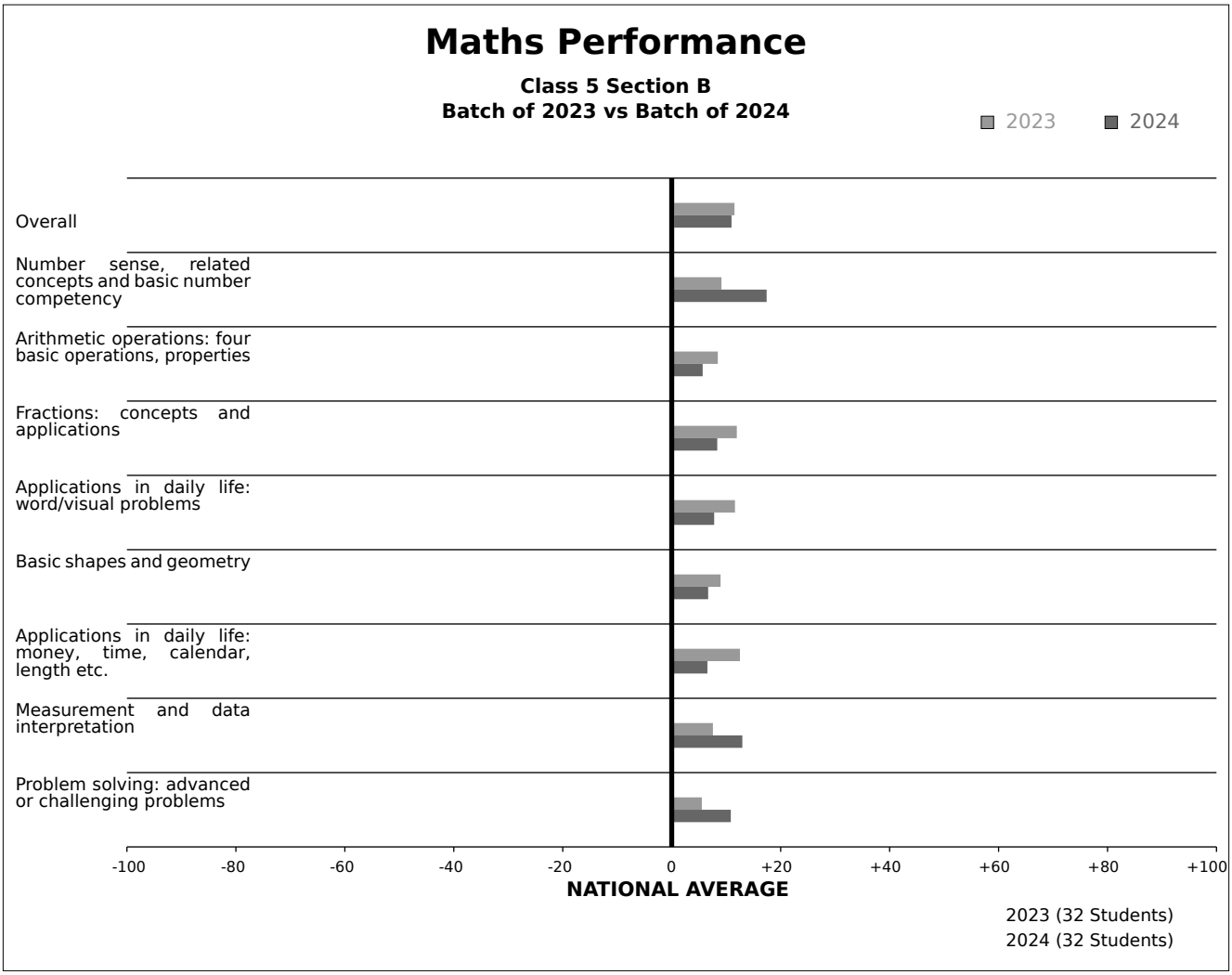
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 5 B Subject: Maths Number of Students: 32

S.No.	Skill	Questions	Class 5 B Performance	Class 5 Performance (School)	Class 5 Performance (National)
			Average	Average	Average
1*	Number sense, related concepts and basic number competency	1,14,21,23,26	73.8%	70.4%	56.3%
2*	Arithmetic operations: four basic operations, properties	5,28,31,32,40	50.0%	58.4%	44.3%
3*	Fractions: concepts and applications	11,13,16,27,37	55.6%	60.8%	47.3%
4*	Applications in daily life: word/visual problems	4,6,7,12,20	67.5%	71.2%	59.7%
5*	Basic shapes and geometry	2,9,17,30,34	61.2%	64.3%	54.6%
6*	Applications in daily life: money, time, calendar, length etc.	3,8,10,25,29	52.5%	60.8%	46.0%
7*	Measurement and data interpretation	19,22,33,35,39	62.5%	61.6%	49.5%
8*	Problem solving: advanced or challenging problems	15,18,24,36,38	53.8%	55.7%	42.9%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Number sense, related concepts and basic number competency	1.	Arithmetic operations: four basic operations, properties
2.	Applications in daily life: word/visual problems	2.	Applications in daily life: money, time, calendar, length etc.

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 30

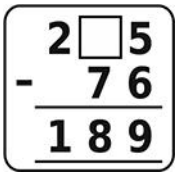
How many square tiles (as shown above) will be needed to cover the rectangular area below?

A 4 **B** 6 **C** 8 **D** 10

Skill: Basic shapes and geometry

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
30	25224	C	28%	40%	Your School	
					All Schools	

Q 31 The subtraction problem shown below is solved correctly.



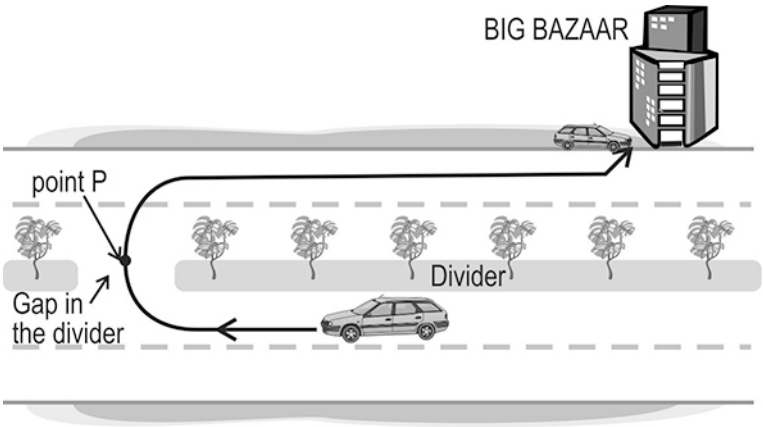
What digit goes into the empty box?

A 1 **B** 4 **C** 5 **D** 6

Skill: Arithmetic operations: four basic operations, properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
31	25224	D	38%	45%	Your School	
					All Schools	

Q 20 Sunil goes to BIG BAZAAR in his car, along the path shown in the picture below.



The length of the path from the gap in the divider (point P) to BIG BAZAAR is 2 km.

Roughly, how much distance should Sunil drive from his current position to reach BIG BAZAAR?

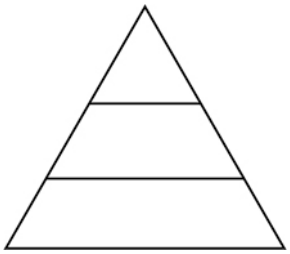
A 1 km **B** 2 km **C** 3 km **D** 4 km

Skill: Applications in daily life: word/visual problems

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
20	25224	C	44%	49%	Your School	
					All Schools	

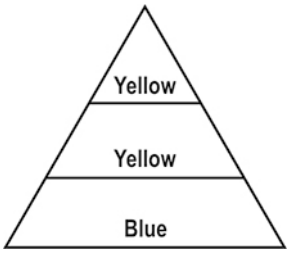
Questions which students of your school found the EASIEST compared to the rest of the country:

Q 24 Given below is a triangle divided into 3 parts.



Anshul has to colour the 3 parts with two colours - yellow and blue. Both the colours have to be used. The same colour cannot be used in all the 3 parts.

(Hint: One way of colouring it is this - first two parts yellow and the third part blue, as shown below.)



In how many ways can he colour the 3 parts of the triangle?

A 12 **B** 6 **C** 3 **D** 2

Skill: Problem solving: advanced or challenging problems

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
24	25224	B	66%	34%	Your School	
					All Schools	

Q 4 Manya made candles and packed them in boxes such that each box had 6 candles. 2 candles remained after packing.

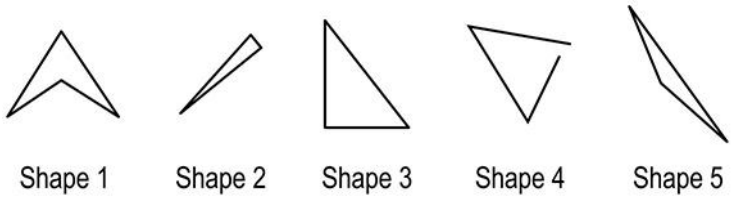
Which of the following could be the number of candles she made?

A 16 **B** 18 **C** 20 **D** 22

Skill: Applications in daily life: word/visual problems

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
4	25224	C	78%	51%	Your School	
					All Schools	

Q 9 Which of these is a triangle?



A only shape 3
B only shapes 3 and 4
C only shapes 1, 3 and 4
D only shapes 2, 3 and 5

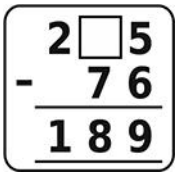
Skill: Basic shapes and geometry

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
9	25224	D	75%	50%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 31 The subtraction problem shown below is solved correctly.



What digit goes into the empty box?

A 1 **B** 4 **C** 5 **D** 6

This question tests whether students have understood the subtraction-WITH-borrow algorithm. Students were expected TO identify the two occurrences of borrowing Ã”Ã‡Ã´ FROM the hundreds TO the tens place, AND FROM the tens TO the ones place. SOME students have only observed the former, thereby finding that TO subtract 7 tens AND get 8 tens IN the answer, the missing digit must have been 5 (FROM 15 tens). Draw studentsÃ”Ã‡Ã- attention TO the ones COLUMN AND the tens COLUMN Ã”Ã‡Ã´ HELP them see that borrowing has occurred IN BOTH places AND thus identify the missing digit AS 6 (7 tens have been subtracted FROM 15 tens, but 1 ten has already been borrowed IN ORDER TO be able TO subtract 6 ones FROM 15 ones). Students may also realise that adding 189 AND 76 would HELP them arrive AT the solution.

Papercode: 25224, Question no: 31					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	16%	3%	44%	38%	32
National	15%	10%	28%	45%	14049

Q 32 89 × 7 = 623
89 × 9 = 801

Which of these is 89 × 79?
(Note: You need not calculate.)

A 623 + 801 **B** 623 × 801 **C** 623 + 8010 **D** 6230 + 801

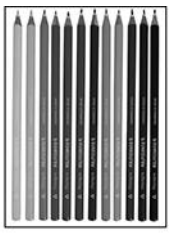
This question was designed TO test students' understanding of the multiplication algorithm. When multiplying 89 by 79, students are used to multiplying 89 by 9 and then 89 by 7 and then adding these - they may not have noticed that when multiplying 89 by 7, they write the product aligned in a certain way. Hence many of them have answered A. Help them understand that 89 Ã— 79 is the same as 89 Ã— 70 + 89 Ã— 9. Also help them see how to arrive at 89 Ã— 70 from 89 Ã— 7. Relate these to the way they align the products in the multiplication algorithm and help them see why it works.

Papercode: 25224, Question no: 32					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	38%	16%	9%	38%	32
National	26%	26%	17%	29%	14049

Q 10 Archana bought TWO of the items shown below.

Rs 44

Rs 15



Rs 27

If she paid with a 100-rupee note, what is the MAXIMUM amount of change she could have received?

A Rs 29 **B** Rs 56 **C** Rs 58 **D** Rs 85

This question tests whether the student understands the relation between the bill value and the amount of change received. Students were expected to realise that in order to get the maximum amount of change, the bill value should be the least, and thus the two least expensive items should be purchased. Many students have associated the word Ã”Ã‡Â£maximumÃ”Ã‡Ã˜ with the maximum priced items and thus subtracted the sum of 44 and 27 from 100 to choose option A. Other students have understood the least expensive items should be purchased, but have missed the word Ã”Ã‡Â£twoÃ”Ã‡Ã˜ and subtracted only 15 from 100. Help students understand that the smaller the number subtracted, the larger the answer, and vice-versa. Allow them to observe this by engaging them in shopkeeper-customer role play in the classroom.

Papercode: 25224, Question no: 10					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	31%	6%	50%	13%	32
National	35%	9%	35%	18%	14049

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Arithmetic operations: four basic operations, properties (50%)

Q 1 If a number is divided by 9, what is the greatest remainder it can leave?

A 10 **B** 9 **C** 8 **D** 1

Explanation:

C: The remainder when a number is divided by 9 is what remains when we have taken away as many 9s as possible from the number. So the remainder has to be less than 9. If we have a remainder 9 or greater, it means that we can still take away 9. So the greatest remainder possible when a number is divided by 9 is 8.

Q 2 In her rough work, Janu added 2389 and 740 as shown here.

Is it okay to write the number 740 this way in the addition?

(*Hint: You do NOT need to calculate*)

A Yes, because 0 has no value so 74 is the same as 740.
B No, because the answer will be wrong if 0 is not written.
C No, because the digits 7 and 4 have NOT been arranged according to their value.
D Yes, because 7 is added with the hundred's digit and 4 with the ten's digit.

Explanation:

D: 740 is 7 hundreds and 4 tens. So adding 740 to 2389 is the same as adding 7 hundreds and 4 tens. Janu has added 7 to the hundred's digit in 2389 and 4 to the ten's digit, and so she is correct. She has to add 0 to the ones digit of 2389, as there are 0 ones in 740. So it is alright even if she doesn't write the 0, as long as she is adding the *7 to the hundred's digit and 4 to the tens digit* . Also 74 is NOT the same as 740, just as 1 is not the same as 10 or even 100.

Q 3 Balbir, Gunjan and Alya have written 3 number sentences.

Balbir:	**815 = 800 + 15**
Gunjan:	**800 + 15 = 815**
Alya:	**800 + 10 + 5 = 815**

Whose number sentence is correct?

A only Gunjan's
B only Alya's
C only Gunjan's and Alya's
D All of them are correct.

Explanation:

D: The equal sign tells us that the expressions on both its sides have the same value. Here the expressions on both sides of the = sign in all the 3 number sentences have the same value, and so all the three sentences are correct. "=" does not mean that what comes to its right has to be the answer to the problem on its left. When we write 12 = 7 + 5, we are just saying that 12 is equal in value to 7 + 5.

Q 4 An airline allows 20 kg baggage for each person. Any extra weight above that is charged at Rs. 70 per kg.

i.e. if one carries 1 kg extra, one will have to pay Rs. 70.

Anita's luggage weighed 25 kg. What amount will she have to pay for the extra luggage that she carried?

A Rs. 70 **B** Rs. 350 **C** Rs. 1400 **D** Rs. 1750

Explanation:

B: Anita does not have to pay anything for the first 20 kg of her luggage. She has to pay only for the extra luggage that she carries over 20 kg. She carries 25 - 20 = 5 kg extra luggage. For each kilogram of extra luggage, she has to pay Rs. 70. So for 5 kg she has to pay 5 × 70 = Rs. 350.

Q 5 □

÷ 100 = 50

The number in the empty box is

A 2 **B** 150 **C** 500 **D** 5000

Explanation:

D: Here we need to find out what number when divided by 100, will give an answer 50. Clearly 100 × 50 = 5000 gives an answer 50 when divided by 100. Note that the question is NOT asking for 100 ÷ 50.

Q 6 The sum of 2 numbers is 250. One of them is greater than 150 .

Which of these is definitely true about the other number?

A It is equal to 100.
B It has to be less than 100.
C It has to be greater than 100.
D It has to be a number between 150 and 250.

Explanation:

B: The sum of 2 numbers is 250- this is the same as saying that if you have some marbles, and I have some number of marbles, the total number of marbles is 250. Now if you have 150 marbles and the total number is 250, how many will I have? 250-150 or 100. So if you have more than 150 marbles and the total number is 250, how many will I have? It has to be less than 100, because if I had more, then the total would be more than 250! So the answer is B.

Q 7 What number should come in the empty box to make the statement true?

55 + 300 - 60 + 45 = □ + 240

A 100 **B** 240 **C** 300 **D** 400

Explanation:

A: Some problems can be solved easily by grouping the numbers conveniently and rearranging them. In the given question, it can easily be seen that 300 - 60 = 240. So the problem can be written as 55 + 300 - 40 + 45 which simplifies to 55 + 240 + 45. This can be rearranged as 55 + 45 + 240 which is equal to 100 + 240. The equal sign means that the expressions on both its sides are equal. Comparing this with the expression on the right side, we see that the number that should go in to the box is 55 + 45 = 100.

Q 8 76 × 2 = 152 and 76 × 8 = 608. Which of these is equal to 76 × 28?

(Note: You need not actually multiply or add.)

A 152 + 608 **B** 1520 + 608 **C** 6080 + 152 **D** 152 × 608

Explanation:

B: 76 × 28 is 28 times 76, that is 20 times 76 + 8 times 76. We are given that 76 × 2 = 152, so 76 × 20 is 1520. We are also given 76 × 8 = 608. So 20 times 76 + 8 times 76 = 1520 + 608.

Skill: Applications in daily life: money, time, calendar, length etc. (53%)

Q 1 Every postage stamp has some value. The value of the first stamp shown below is 20 paise. The value of the second one is Rs. 4.

Rahul needs to paste a stamp of Rs. 15 on something he has to send by post. Which of the following should he select?

A

A.

B

B.

C

C.

D

D.

Explanation:

D: Note that in the four rupee stamp, the value is written as 400. 1 rupee = 100 paise and Rs. 4 = 400 paise. So the value of the stamp is written on it in paise. Rs. 15 = 15 × 100 = 1500 paise, and so the value of the stamp in option D is Rs.15. Note the *P.* on the stamp in option A. The value of the stamp in option A is 15 paise and not Rs. 15.

Q 2 2 nd April, 2005 was Saturday. Which day of the week would 22 nd April, 2005 have been?

A Thursday **B** Friday **C** Saturday **D** Sunday

Explanation:

B: It is given that 2nd April was a Saturday. Can you count and see when the next Saturday will be? 9th April, right! When would the next Saturday be? After the 9th of April, it would be 16th April is it not? Do you see a pattern? We get the next Saturday by adding 7 to the current date. This is true for any day. **E.g.:** If today is Monday, the date on which next Monday would fall is today's date + 7. So we can say that 9th , 16th and 23rd are all Saturdays. Hence, 22nd April would be a Friday.

Q 3 Prakash stood on the weighing machine with his daughter and this is what it showed.

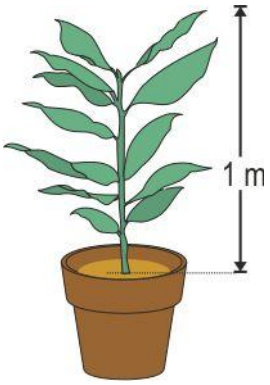
Which of these could be Prakash's weight?

A 25 kg **B** 55 kg **C** 70 kg **D** 75 kg

Explanation:

B: The weight of Prakash and his daughter together is 70 kg. That is Prakash's weight + his daughter's weight is 70 kg. So Prakash alone has to be less than 70 kg. An average adult weighs more than 25 kg. So A cannot be the answer. So Prakash could weigh 55kg.

Q 4 How tall would this plant be if it grew 20 cm more in height?



A 12 m **B** 120 cm **C** 21 cm **D** 12 cm

Explanation:

B: The length of the plant is given to be 1 m. Now 1 m is equal to 100 cm. If the plant grows by another 20 cm, it's height would be 100 + 20 = 120 cm. We cannot add 2 quantities if the units are not the same, that is we cannot add 1 *m* and 20 *cm* , unless we express both in the same units, m or cm.

Q 5 Shown below is a part of a train ticket of a train from Lokmanya Tilak Terminus, Mumbai to Coimbatore Junction.

IRCTC Ltd, Booked Ticket Printing

IRCTC's e-Ticketing Service
Electronic Reservation Slip

IRCTC

Transaction ID: 0090130616		PNR No.: 8327887718
Train No. & Name: 1013/COIMBATORE EXP		Date of Journey: 11-Apr-2009
Class: 3A	Date of Booking: 27-Jan-2009 01:29:40 PM	Date of Boarding: 11-Apr-2009
From: LOKMANYA TILAK T (LTT)	To: COIMBATORE JN (CBE)	Distance: 1569 KM
Boarding: LOKMANYA TILAK T (LTT)	Resv Upto: COIMBATORE JN (CBE)	Quota: General
Scheduled Departure: 20:30	Total Fare: Rs. 1495.00	Adult: 2 Child: 0

Badri booked the ticket on 27 th January 2009, for travel on 11 th April 2009. Roughly how much in advance did he book the ticket?

A 2 months **B** 2 and a half months **C** 3 and a half months **D** 4 months

Explanation:

B: We need to calculate the time between 27 th January 2009 and 11 th April 2009. We have about 5 days in January, two whole months namely February and March and 11 days in April. That is we have about 2 months and 16 days. That is roughly 2 and a half months.

Q 6 Which of these could weigh around 1 gram?

A

Metal spoon

B

Medicine

C

Plate

D

Book

Explanation:

B: A metal spoon could weigh about 10 grams, a plate could weigh 100 grams and a thick book could weigh 1 kilogram or 1000 grams. A medicine tablet as shown here could weigh 1 gram. 1 gram is a really small amount of weight.

Q 7 Pratixa poured 5 cups of water into an empty bottle. The water level came up to half the height of the bottle as shown here.

If she wants to fill the bottle to the top, how many MORE cups of water will she require?

A exactly 5
B less than 5
C more than 5
D We cannot say for sure.

Explanation:

B: Note that some part of the top half of the bottle is narrower than the bottom half. If we have 2 vessels of the same height, one broad and the other narrow, we know that the narrower vessel holds less water than the broader one.Though the heights of both the halves are the same, the top half will hold less water because part of it is narrower. So Pratixa will need less than 5 cups of water to fill the bottle.

Q 8 Given below is a list of all the activities that Madura has to do every morning before going to school and the time taken for each.

Bathing and dressing: 40 minutes
Packing her schoolbag: 5 minutes
Having breakfast: a quarter of an hour
Walking to the bus stop: 5 minutes

At about what time should she start getting ready if she wants to reach the bus stop by 7:25 every morning?

A 5:45 a.m
B 6:20 a.m
C 6:30 a.m
D 6:45 a.m

Explanation:

B: There are 60 minutes in an hour, and so a quarter of an hour would be $\frac{1}{4}$ of 60 = 15 minutes.

Madura takes 40 + 5 + 15 + 5 = 65 minutes, that is 1 hour and 5 minutes, in all to get ready. She has to reach the bus stop at 7:25 in the morning, so she should get up 1 hour and 5 minutes before 7:25 to get sufficient time to get ready. 1 hour and 5 minutes before 7 : 25 is 6: 20.

Notes